# Coupling-Aware Estimation in CTMT

A Cramér–Rao Efficiency Gap, its Canonical-Correlation Law, and the Calibration Condition for Null-Sector Gain

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**Abstract**

This note establishes the first *supporting pillar* of CTMT as an estimation principle, independent of any physical interpretation. Working in the resolution geometry $\mathcal{G} = (O, g; R, \Sigma)$ with resolved sector $R = \mathrm{Im}(J)$ and null sector $N = R^{\perp_g}$, we compare two unbiased estimators of the parameter $\theta$: the *resolved-only* generalized least squares (GLS) estimator, which uses only the signal-bearing sector, and the *coupling-aware* GLS estimator, which additionally uses the null sector as a correlated noise reference. We prove that the coupling-aware estimator dominates the resolved-only estimator in the Löwner order, strictly whenever the resolved–null covariance coupling $C_{RN}$ is nonzero, and that the efficiency gain is governed exactly by the canonical correlations $\rho_i$ of the whitened coupling: each contributes a leverage factor $\rho_i^2/(1-\rho_i^2)$, and in the just-identified case the ratio of Cramér–Rao confidence-ellipsoid volumes is exactly $\prod_i(1-\rho_i^2)$. Equivalently, half the log-volume reduction equals the resolved–null mutual information, tying this efficiency statement to the information invariant of the static theory. Crucially, the gain is conditional: it requires a *calibrated null baseline*. If the null-sector mean is an unknown nuisance, profiling it out returns the resolved-only information and the gain vanishes. All results are exact statements of standard estimation theory (Cramér–Rao, Gauss–Markov, sufficiency), verified numerically, and hold as a decision-relevant guarantee rather than a physical claim.

## Contents

# 1 Purpose

Papers I–III established that the resolved–null coupling $C_{RN}$ is a genuine, non-redundant invariant of the resolution geometry, and that the null sector, though signal-free on its own, increases the Fisher information about the resolved parameters. That was a statement about *information*. Here we convert it into a statement about *estimators*, which is what a statistician will actually weigh:

> *If you ignore the null sector, how much accuracy do you lose, and under what condition do you lose it?*

The answer is a Cramér–Rao efficiency gap with an exact canonical-correlation law, and a precise calibration condition. Nothing in this note depends on a physical interpretation of CTMT; it is a guarantee about two concrete estimators, provable from standard theory. This is deliberately the most defensible kind of contribution: a correctness-and-efficiency result that a referee cannot dismiss, because it is not an interpretation.

*Non-claim* 1.1 (Estimation, not physics). This note makes no physical claim. It shows that a coupling-aware estimator is more efficient than a resolved-only estimator under a stated calibration condition. Whether a given system's coupling is physically intrinsic is outside the scope of the result (Papers II–III).

# 2 Setup

**Assumption 2.1** (Linear-mean Gaussian observation model)**.** The observation $y \in O$ is Gaussian,

$$y = \begin{pmatrix} y_R \\ y_N \end{pmatrix} \sim \mathcal{N}\left( \begin{pmatrix} \mu_R(\theta) \\ \mu_N \end{pmatrix},\ \Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix} \right), \qquad \Sigma \succ 0,$$

in a $g$-orthonormal frame adapted to $O = R \oplus N$. The mean varies only in the resolved sector: $\mu_R(\theta) = \mu_R^0 + J_R\theta$ with $J_R$ of full column rank ($\operatorname{rank} J_R = \dim\theta =: p \le r := \dim R$), while the null mean $\mu_N$ is constant in $\theta$ (this is exactly $\operatorname{Im}(J) = R$). The Schur complement of the null block is $S_R := \Sigma_R - C_{RN}\Sigma_N^{-1}C_{RN}^\top \succ 0$.

The linear-mean assumption makes every statement exact; the nonlinear case holds locally and asymptotically through the maximum-likelihood estimator, as noted in Section 7.

**Definition 2.2** (The two estimators)**.** The *resolved-only* GLS estimator uses $y_R$ alone,

$$\hat\theta_R = \left(J_R^\top \Sigma_R^{-1} J_R\right)^{-1} J_R^\top \Sigma_R^{-1}(y_R - \mu_R^0),$$

with information $\mathcal{I}_R = J_R^\top \Sigma_R^{-1} J_R$. The *coupling-aware* GLS estimator uses all of $y$,

$$\hat\theta_{\text{full}} = \left(J^\top \Sigma^{-1} J\right)^{-1} J^\top \Sigma^{-1}(y - \mu^0), \qquad J = \begin{pmatrix} J_R \\ 0 \end{pmatrix},$$

with information $\mathcal{I}_{\text{full}} = J^\top \Sigma^{-1} J$.

# 3 The null sector as a calibrated noise reference

The mechanism by which a signal-free sector helps is classical: it is a correlated noise reference. We make this exact.

**Lemma 3.1** (Corrected observation and sufficiency)**.** *Under Assumption 2.1, define the corrected observation*

$$z := y_R - C_{RN}\Sigma_N^{-1}(y_N - \mu_N).$$

*Then $z$ is a sufficient statistic for $\theta$, and*

$$z \sim \mathcal{N}\big(\mu_R(\theta),\ S_R\big), \qquad S_R = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{RN}^\top.$$

*In particular the coupling-aware information equals the information of $z$:*

$$\mathcal{I}_{\text{full}} = J^\top\Sigma^{-1}J = J_R^\top S_R^{-1}J_R.$$

*Proof.* Linearity gives $\mathbb{E}[z] = \mu_R(\theta) - C_{RN}\Sigma_N^{-1}(\mu_N - \mu_N) = \mu_R(\theta)$ and $\mathrm{Cov}(z) = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{RN}^\top - C_{RN}\Sigma_N^{-1}C_{RN}^\top + C_{RN}\Sigma_N^{-1}\Sigma_N\Sigma_N^{-1}C_{RN}^\top = S_R$. Change variables $(y_R, y_N) \mapsto (z, y_N)$; the Jacobian is unimodular and $y_N$ has a $\theta$-free law (mean $\mu_N$ known and constant, covariance $\Sigma_N$), so the joint density factors as $p(z \mid \theta)\,p(y_N)$ and the $\theta$-dependence resides entirely in $z$: $z$ is sufficient. Its Gaussian information is $J_R^\top S_R^{-1}J_R$. The identity $J^\top\Sigma^{-1}J = J_R^\top(\Sigma^{-1})_{RR}J_R = J_R^\top S_R^{-1}J_R$ follows from block inversion, $(\Sigma^{-1})_{RR} = S_R^{-1}$. $\square$

*Remark* 3.2. The null sector carries no signal ($\mu_N$ is $\theta$-free), yet observing it lets one subtract the correlated component of the resolved-sector noise, leaving the smaller effective noise $S_R \preceq \Sigma_R$. This is the standard control-variate / noise-reference mechanism, and it is the precise, defensible content of "the null sector feeds resolved reconstruction."

## 4 The efficiency gap

**Theorem 4.1** (Coupling-aware dominance)**.** *Under Assumption 2.1, both estimators of Definition 2.2 are unbiased and achieve their Cramér–Rao bounds, with covariances*

$$V_R = \mathcal{I}_R^{-1} = \big(J_R^\top\Sigma_R^{-1}J_R\big)^{-1}, \qquad V_{\text{full}} = \mathcal{I}_{\text{full}}^{-1} = \big(J_R^\top S_R^{-1}J_R\big)^{-1}.$$

*Then*

$$\boxed{\mathcal{I}_{\text{full}}\ \succeq\ \mathcal{I}_R \qquad \Longleftrightarrow \qquad V_{\text{full}}\ \preceq\ V_R,}$$

*with equality if and only if $C_{RN} = 0$. When $C_{RN} \neq 0$ the coupling-aware estimator is strictly more efficient: $V_R - V_{\text{full}} \succeq 0$ is nonzero.*

*Proof.* For a linear-mean Gaussian model the GLS estimator is the maximum-likelihood estimator; it is unbiased and attains the Cramér–Rao bound, so its covariance equals the inverse information (Gauss–Markov / Aitken) [1, 2]. By Lemma 3.1, $\mathcal{I}_{\text{full}} = J_R^\top S_R^{-1}J_R$. Since $C_{RN}\Sigma_N^{-1}C_{RN}^\top \succeq 0$ we have $S_R \preceq \Sigma_R$, hence $S_R^{-1} \succeq \Sigma_R^{-1}$ (operator inversion reverses the Löwner order on the positive-definite cone), and therefore $J_R^\top S_R^{-1}J_R \succeq J_R^\top\Sigma_R^{-1}J_R$, i.e. $\mathcal{I}_{\text{full}} \succeq \mathcal{I}_R$. Inverting reverses the order: $V_{\text{full}} \preceq V_R$. Equality holds iff $S_R = \Sigma_R$, i.e. iff $C_{RN}\Sigma_N^{-1}C_{RN}^\top = 0$; as $\Sigma_N \succ 0$ this is iff $C_{RN} = 0$. $\square$

## 5 The canonical-correlation law

We now quantify the gap by the invariants of the coupling.

**Definition 5.1** (Whitened coupling and canonical correlations)**.** The whitened coupling is $\widetilde{C} := \Sigma_R^{-1/2}C_{RN}\Sigma_N^{-1/2}$; its singular values $\rho_1 \geq \cdots \geq \rho_k > 0$ (with $k = \operatorname{rank} C_{RN}$) are the canonical correlations of the resolved and null sectors, and satisfy $\rho_i < 1$ because $\Sigma \succ 0$.

**Theorem 5.2** (Efficiency gain in canonical correlations)**.** *Write $M := \Sigma_R^{-1/2} J_R$ and diagonalize $\widetilde{C}\widetilde{C}^\top = U\,\mathrm{diag}(\rho_i^2)\,U^\top$. Then the information gain is*

$$\mathcal{I}_{\text{full}} - \mathcal{I}_R = M^\top U\,\mathrm{diag}\left(\frac{\rho_i^2}{1-\rho_i^2}\right) U^\top M.$$

*Each canonical correlation $\rho_i$ contributes a leverage factor $\rho_i^2/(1-\rho_i^2)$, which diverges as $\rho_i \to 1$. Moreover, in the just-identified case $p = r$ (so $J_R$ and $M$ are invertible),*

$$\boxed{\frac{\det V_{\text{full}}}{\det V_R} = \det\big(I - \widetilde{C}\widetilde{C}^\top\big) = \prod_{i=1}^{k}(1-\rho_i^2).}$$

*Proof.* From $S_R = \Sigma_R^{1/2}(I-\widetilde{C}\widetilde{C}^\top)\Sigma_R^{1/2}$ we get $S_R^{-1} = \Sigma_R^{-1/2}(I-\widetilde{C}\widetilde{C}^\top)^{-1}\Sigma_R^{-1/2}$, so $\mathcal{I}_{\text{full}} = M^\top(I-\widetilde{C}\widetilde{C}^\top)^{-1}M$ and $\mathcal{I}_R = M^\top M$. Subtracting and using $(I-\widetilde{C}\widetilde{C}^\top)^{-1} - I = U\,\mathrm{diag}(\rho_i^2/(1-\rho_i^2))\,U^\top$ gives the gain formula. For $p = r$, $V_R = M^{-1}M^{-\top}$ and $V_{\text{full}} = M^{-1}(I - \widetilde{C}\widetilde{C}^\top)M^{-\top}$, so $\det V_{\text{full}}/\det V_R = \det(I - \widetilde{C}\widetilde{C}^\top) = \prod_i(1-\rho_i^2)$. □

**Corollary 5.3** (Efficiency gain equals the sector mutual information)**.** *In the just-identified case,*

$$\tfrac{1}{2}\log\frac{\det V_R}{\det V_{\text{full}}} = -\tfrac{1}{2}\sum_i \log(1-\rho_i^2) = I(y_R; y_N),$$

*the resolved–null mutual information of the static theory (Paper III). The log-volume reduction of the Cramér–Rao confidence ellipsoid is exactly twice the information the null sector shares with the resolved sector.*

*Proof.* Combine Theorem 5.2 with the Gaussian identity $I(y_R; y_N) = -\frac{1}{2}\log\det(I - \widetilde{C}\widetilde{C}^\top)$ [3]. □

This is the internal-coherence payoff: the estimation efficiency gained by coupling-awareness is not merely bounded by, but *equal to*, the information invariant already identified in the static geometry.

# 6 The calibration condition (when the gain is real)

The gain of Theorems 4.1–5.2 is not unconditional. It depends on whether the null-sector baseline $\mu_N$ is known. This is the honest hinge of the whole pillar and the part a careful statistician will test.

**Theorem 6.1** (Calibration dichotomy)**.** *Consider the null mean $\mu_N$ as follows.*

1. Calibrated null baseline ($\mu_N$ known). *The effective information for $\theta$ is $\mathcal{I}_{\text{full}} = J_R^\top S_R^{-1} J_R \succeq \mathcal{I}_R$, with the canonical-correlation gain of Theorem 5.2.*
2. Uncalibrated null baseline ($\mu_N$ an unknown, unconstrained nuisance). *Profiling out $\mu_N$ returns the resolved-only information exactly,*

$$\mathcal{I}_{\theta|\mu_N} = J_R^\top\big(P - QS^{-1}Q^\top\big)J_R = J_R^\top \Sigma_R^{-1} J_R = \mathcal{I}_R,$$

*where $\Sigma^{-1} = \begin{pmatrix} P & Q \\ Q^\top & S \end{pmatrix}$. The null-sector gain vanishes.*

*Proof.* Case (1) is Theorem 4.1. For (2), the joint mean is $(\mu_R^0 + J_R\theta,\ \mu_N)$ with free $\mu_N$; the joint Fisher information in $(\theta, \mu_N)$ is $\begin{pmatrix} J_R^\top P J_R & J_R^\top Q \\ Q^\top J_R & S \end{pmatrix}$. The information for $\theta$ after profiling the nuisance $\mu_N$ is the Schur complement $J_R^\top(P - QS^{-1}Q^\top)J_R$. By block inversion, $P - QS^{-1}Q^\top = \Sigma_R^{-1}$, giving $\mathcal{I}_R$. Verified numerically. □

*Remark* 6.2 (Operational meaning). The null sector helps only as a *calibrated* noise reference. If its baseline is pinned down—by design, prior calibration, or a known physical zero—the coupling-aware estimator delivers the full canonical-correlation gain. If the null baseline is left free, it must be estimated, and doing so consumes exactly the advantage the correlation would have provided. This turns CTMT's null sector into a concrete design recommendation: calibrate the unresolved channels, then use them as references.

## 7 Scope and non-claims

*Non-claim* 7.1 (Linear-mean, then local/asymptotic). All exact statements assume a linear mean (Assumption 2.1). For a nonlinear mean the results hold locally at $\theta$ through the Fisher information $J^\top \Sigma^{-1} J$ and asymptotically through the efficiency of the maximum-likelihood estimator; the canonical-correlation law then describes the local confidence ellipsoid.

*Non-claim* 7.2 (Requires calibration). The efficiency gain requires a calibrated null baseline (Theorem 6.1); with an uncalibrated null mean there is no gain.

*Non-claim* 7.3 (Known covariance). $\Sigma$ is taken as known (or well estimated). Covariance-estimation error propagates into the realised gain and is a separate, standard concern.

*Non-claim* 7.4 (Efficiency, not truth). The result says the coupling-aware estimator is more efficient, not that the resolution geometry is physically correct (Non-claim 1.1).

## 8 Operational reading and a falsifiable check

The pillar yields a concrete, testable guarantee. On any linear-mean Gaussian chart with a calibrated null baseline:

- the confidence-ellipsoid volume of the coupling-aware estimator is exactly $\prod_i (1 - \rho_i^2)$ times that of the resolved-only estimator (Theorem 5.2);
- the log-volume saving equals the sector mutual information (Corollary 5.3);
- ignoring the null sector, or leaving its baseline uncalibrated, forfeits this saving (Theorem 6.1).

These are exact identities, so they are trivially checkable on synthetic data (as done in preparing this note, matching to numerical and Monte-Carlo precision) and directly usable as a design principle: a chart with large canonical correlations and a calibrated null baseline should show the predicted variance reduction, and its absence falsifies the calibration assumption for that chart.

## 9 Conclusion

CTMT's static geometry already proved that the null sector carries information about the resolved parameters. This note converts that into the form a statistician will accept as non-trivial: a Cramér–Rao efficiency guarantee for a concrete estimator, quantified exactly by the coupling canonical correlations, equal to the sector mutual information, and valid precisely when the null baseline is calibrated. It is a correctness-and-efficiency result of standard estimation theory, independent of any physical interpretation. As a supporting pillar it says something true and useful now: *treating the resolved–null coupling as a first-class object yields a strictly more efficient estimator, by a computable amount, whenever the unresolved channels are calibrated and correlated with the signal.* That is a defensible claim on its own terms, and it stands whether or not the geometry is ever shown to be physical.

## References


[1] C. R. Rao, *Linear Statistical Inference and Its Applications*, 2nd ed., Wiley, 1973.

[2] E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed., Springer, 1998.

[3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.

[5] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, American Mathematical Society, 1982.